

DHANANAJAY KUMAR SINGH

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PROFESSIONAL SUMMARY

Master of Computer Applications (MCA) graduate with practical experience in Python, Flask, Machine Learning, and Computer Vision. Built AI-powered applications using YOLOv8, OpenCV, MediaPipe, ByteTrack, and FFmpeg for video processing, object detection, and sports analytics. Proficient in developing full-stack web applications, RESTful APIs, and deploying scalable solutions using Git, Docker, and Render. Passionate about building innovative AI-driven software solutions.

TECHNICAL SKILLS

Languages & Databases: C, Python, SQL

Web Development (Frontend & Backend): HTML, CSS, Bootstrap, Flask, Django

Machine Learning: Scikit-learn, Feature Engineering, Model Evaluation

Computer Vision: OpenCV, YOLOv8, MediaPipe Pose

Tools & Platforms: Git, GitHub, Docker, VS Code, Render

ACADEMIC PROJECTS

Overlay Studio | Python, Flask, Rembg (ISNet ONNX), FFmpeg, MoviePy, REST APIs

- Engineered a full-stack Flask application combining AI-powered background removal with interactive video overlay and trimming workflows.
- Integrated rembg with the ISNet ONNX model for automated image segmentation, transparent cutout generation, and customizable background compositing.
- Developed an FFmpeg/MoviePy video processing pipeline supporting logo positioning, scaling, rotation, timeline-based overlays, entrance animations, trimming, and configurable output quality.
- Designed an iframe-isolated frontend architecture and REST APIs for media upload, AI processing, preview generation, progress tracking, video cutting, and output delivery.
- Implemented memory-efficient media handling using RAM-based image caching and Linux /dev/shm temporary storage, with automated cleanup and production deployment support.

Video Watermark Remover | Python, Flask, OpenCV, FFmpeg

- Developed a full-stack web application for detecting and removing watermarks from user-uploaded videos (MP4, AVI, MOV, MKV, WEBM) with support for files up to 50 MB.
- Implemented four configurable watermark removal techniques: AI-based inpainting (OpenCV TELEA), Gaussian blur, pixelation, and black-box masking.
- Integrated FFmpeg for video encoding and to preserve the original audio stream during processing, ensuring minimal quality loss.
- Built a responsive web interface with real-time upload, processing, and download functionality.

20M Dash Speed Analysis | Python, Flask, YOLOv8, ByteTrack, OpenCV, PyTorch

- Developed an AI-powered sprint analysis application that estimates runner speed and running form from 20-meter dash videos using YOLOv8 object detection and ByteTrack multi-object tracking.
- Built an asynchronous Flask + SSE processing pipeline with pixel-to-meter calibration to generate annotated videos and detailed HTML/JSON performance reports.

EDUCATION

Master of Computer Applications (MCA) – LNCT Group of Colleges, Bhopal | 2026 | CGPA: 7.93/10

Bachelor of Computer Applications (BCA) – Magadh University, Bodh Gaya, Bihar | 2024 | Percentage: 70.67%

CERTIFICATIONS

- Internship on AI — Microsoft Azure, Microsoft initiative implemented by Edunet Foundation
- AWS Academy Cloud Foundation — Issued by Amazon Web Services
- YPSILON IT Solutions Private Limited — Onsite training at LNCT Group of Colleges, Bhopal, in Python & Django